64 nodes on Frontera. 8 on Vista.

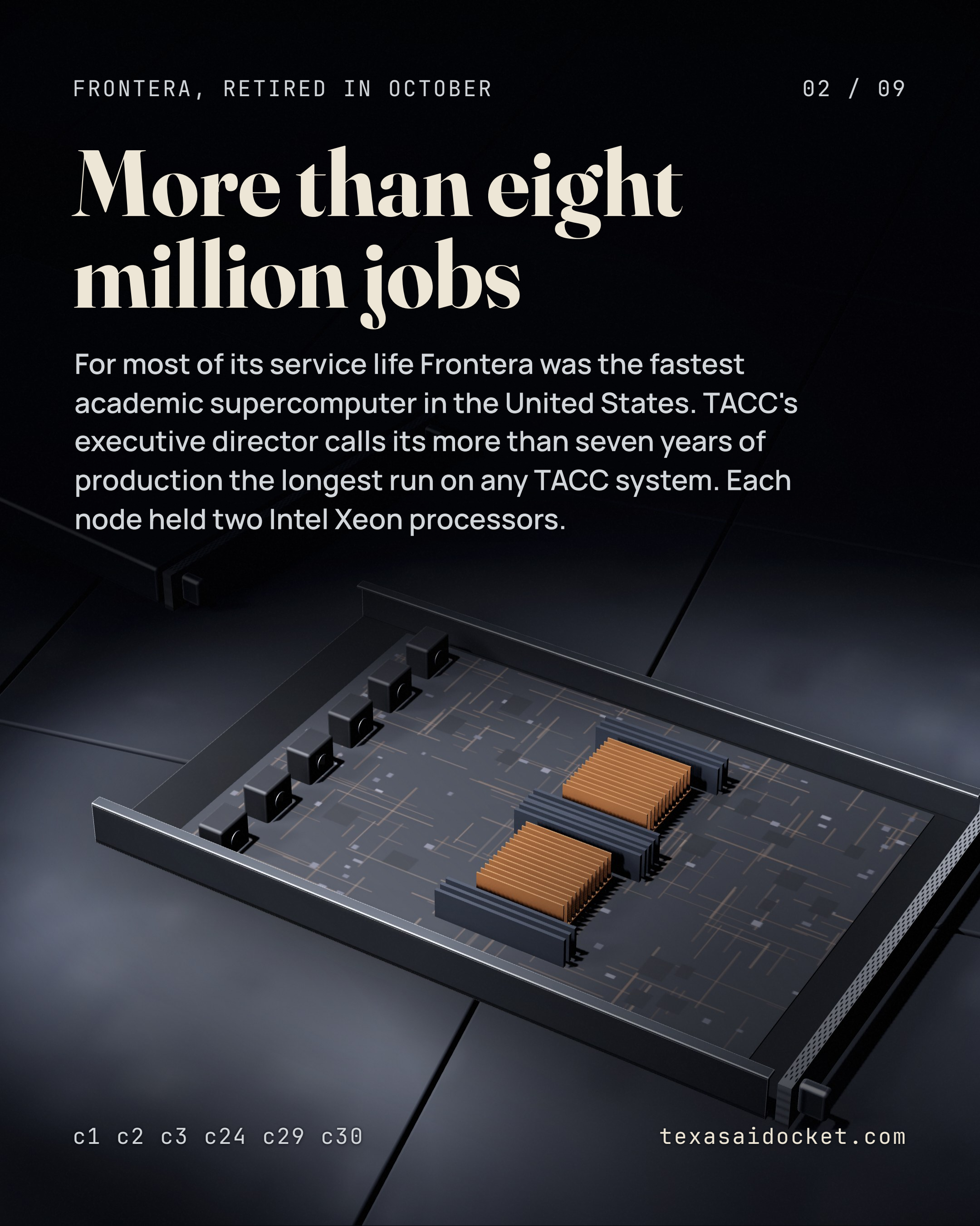
Jose Rizo-Rey's lab at UT Southwestern Medical Center typically ran a simulation on 64 CPU nodes of TACC's Frontera. He says eight GPU nodes of Vista gave "almost the same speed." One tray per node, drawn to illustrate.



c7 c8 c9 c28

More than eight million jobs

For most of its service life Frontera was the fastest academic supercomputer in the United States. TACC's executive director calls its more than seven years of production the longest run on any TACC system. Each node held two Intel Xeon processors.



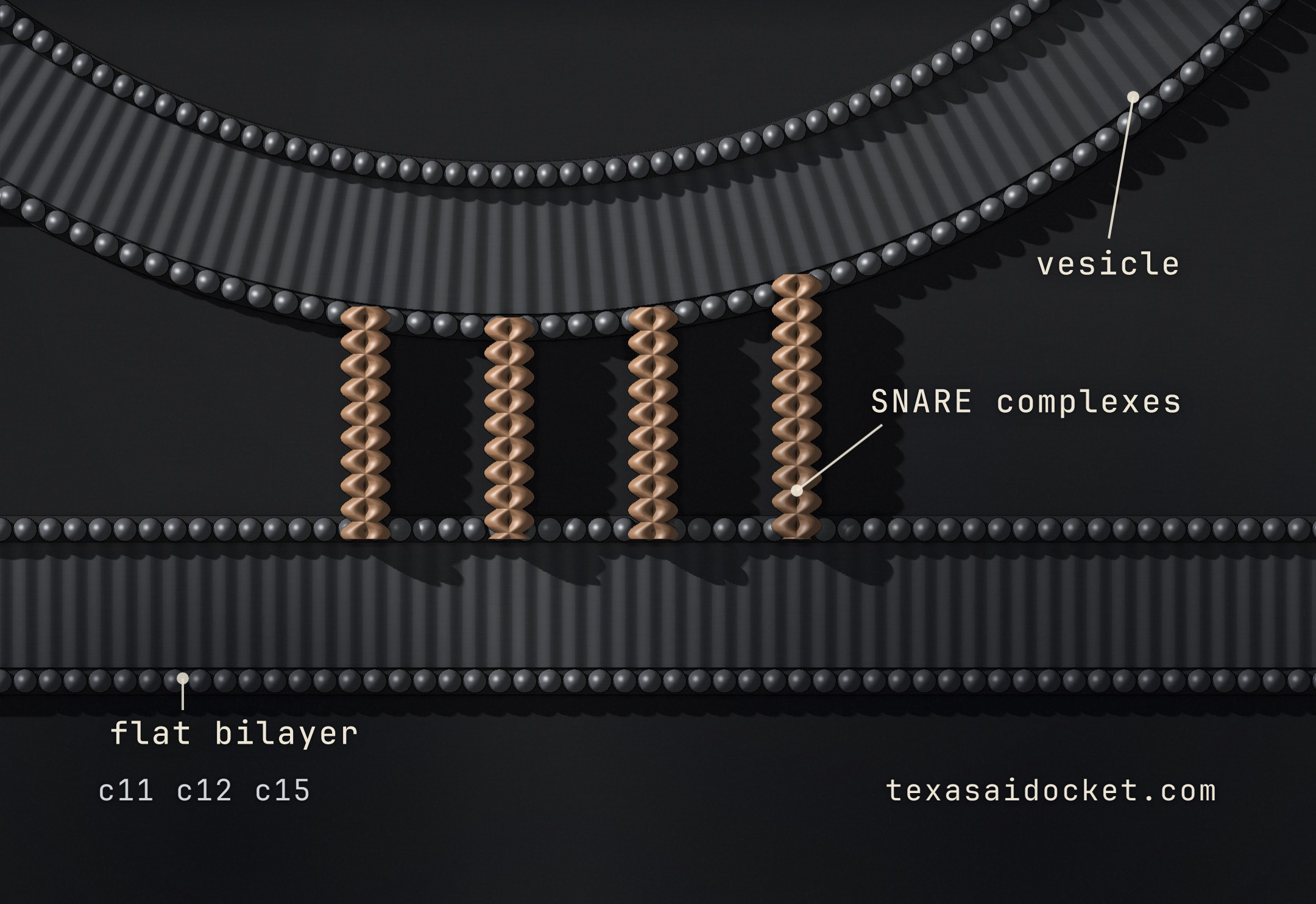
64 CPU nodes against 8 GPU nodes

On 64 Frontera nodes the 5.3 million atom simulation typically ran at 43 nanoseconds a day. Rizo-Rey says eight Vista GPU nodes gave "almost the same speed." His account, not a benchmark. One tray per node.



A vesicle, a bilayer, four SNARE complexes

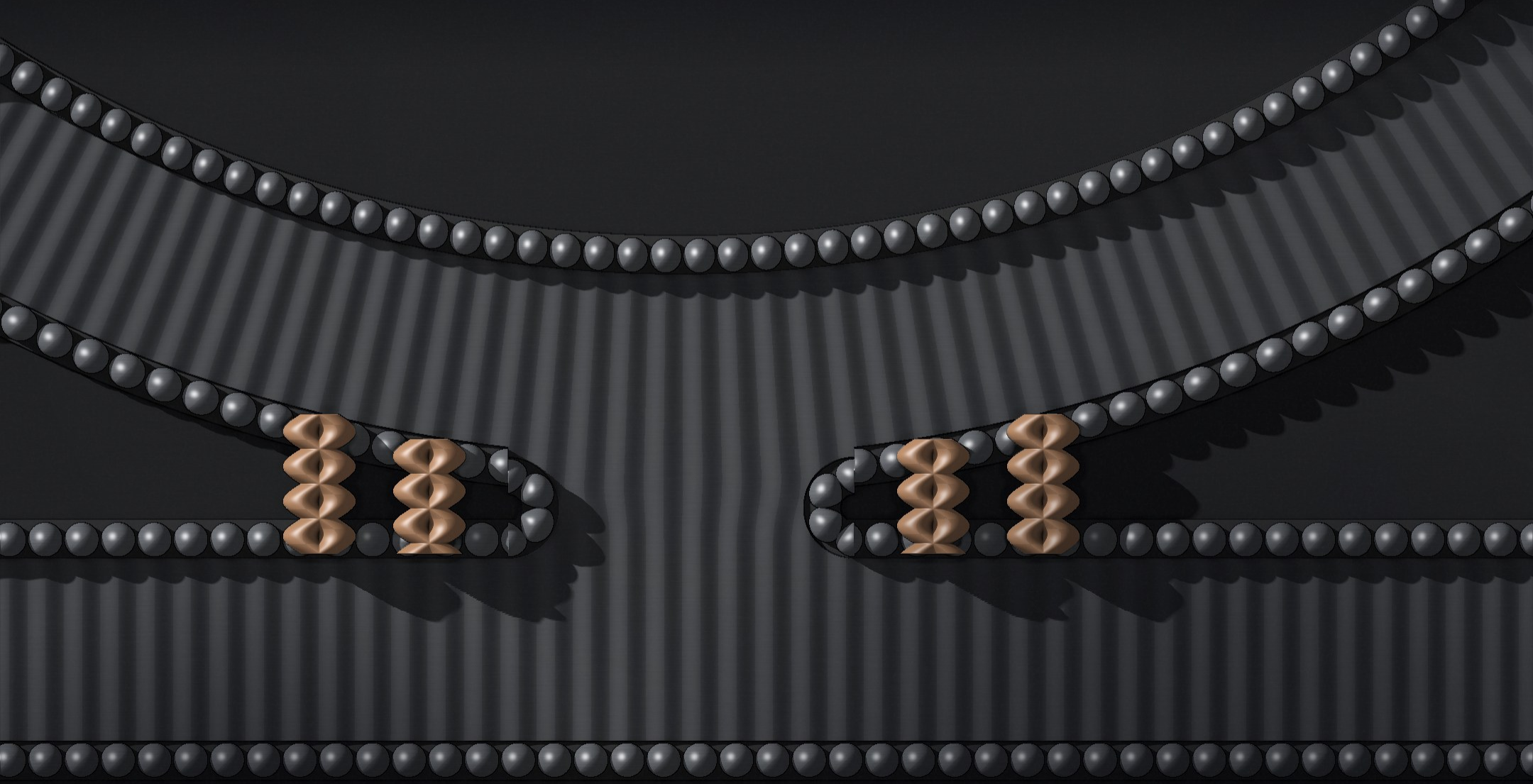
Rizo-Rey's simulations contain a vesicle, a flat bilayer and four SNARE protein complexes. The SNAREs let vesicles fuse with their target membranes. His research focuses on synaptic transmission. Drawn to illustrate.



c11 c12 c15

SNAREs act as a local detergent

Rizo-Rey calls it the local detergent model of membrane fusion. In it the SNAREs catalyze mixing of the hydrophobic interiors of the two membranes. Drawn to illustrate, not a frame from the simulation.



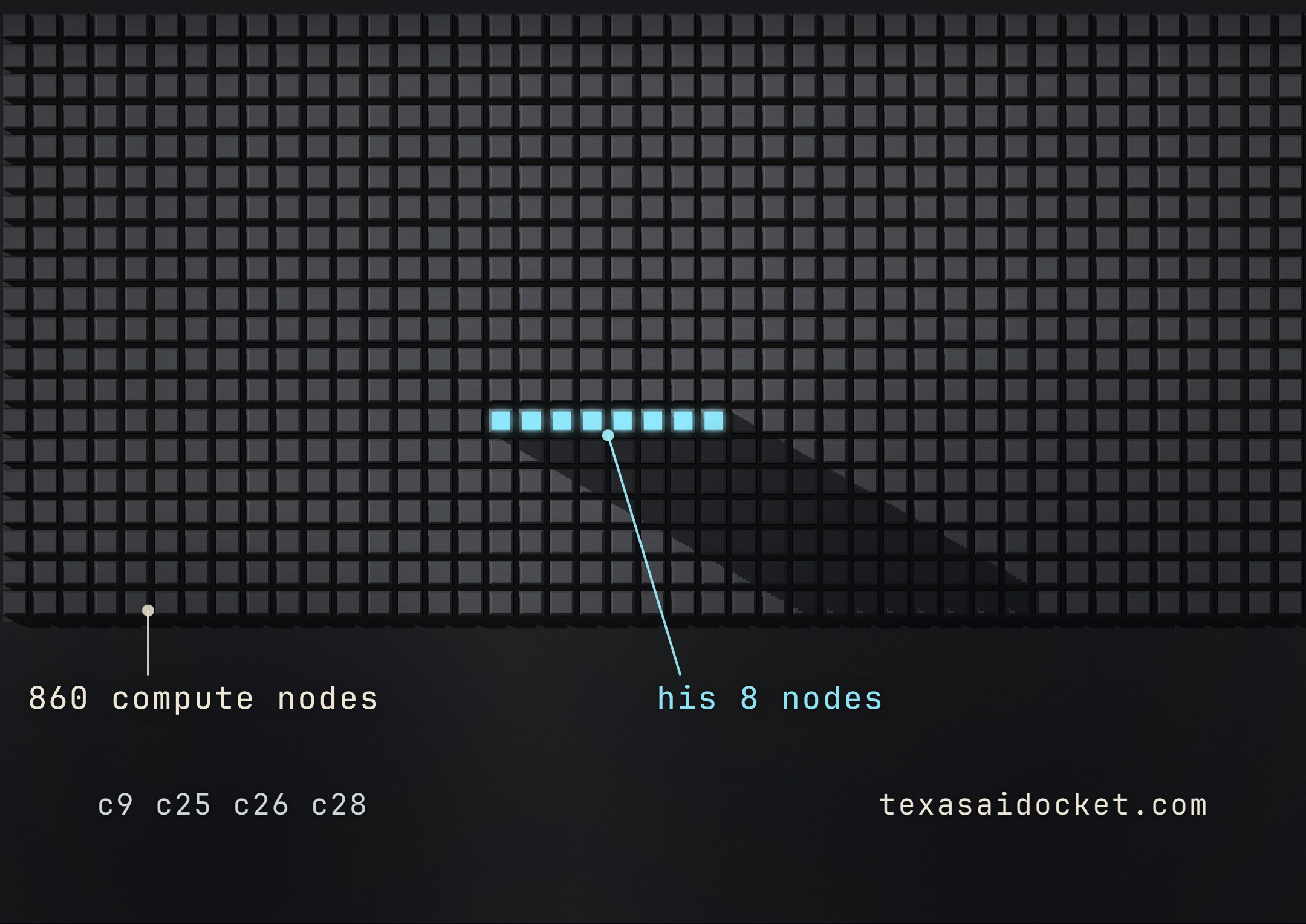
The speed took a lot of tuning

Rizo-Rey says reaching it on Vista "required a lot of optimizations with GROMACS, with the help of people here at TACC." GROMACS is free and open source software for molecular dynamics.



Rizo-Rey's 8 nodes, in Vista's 860

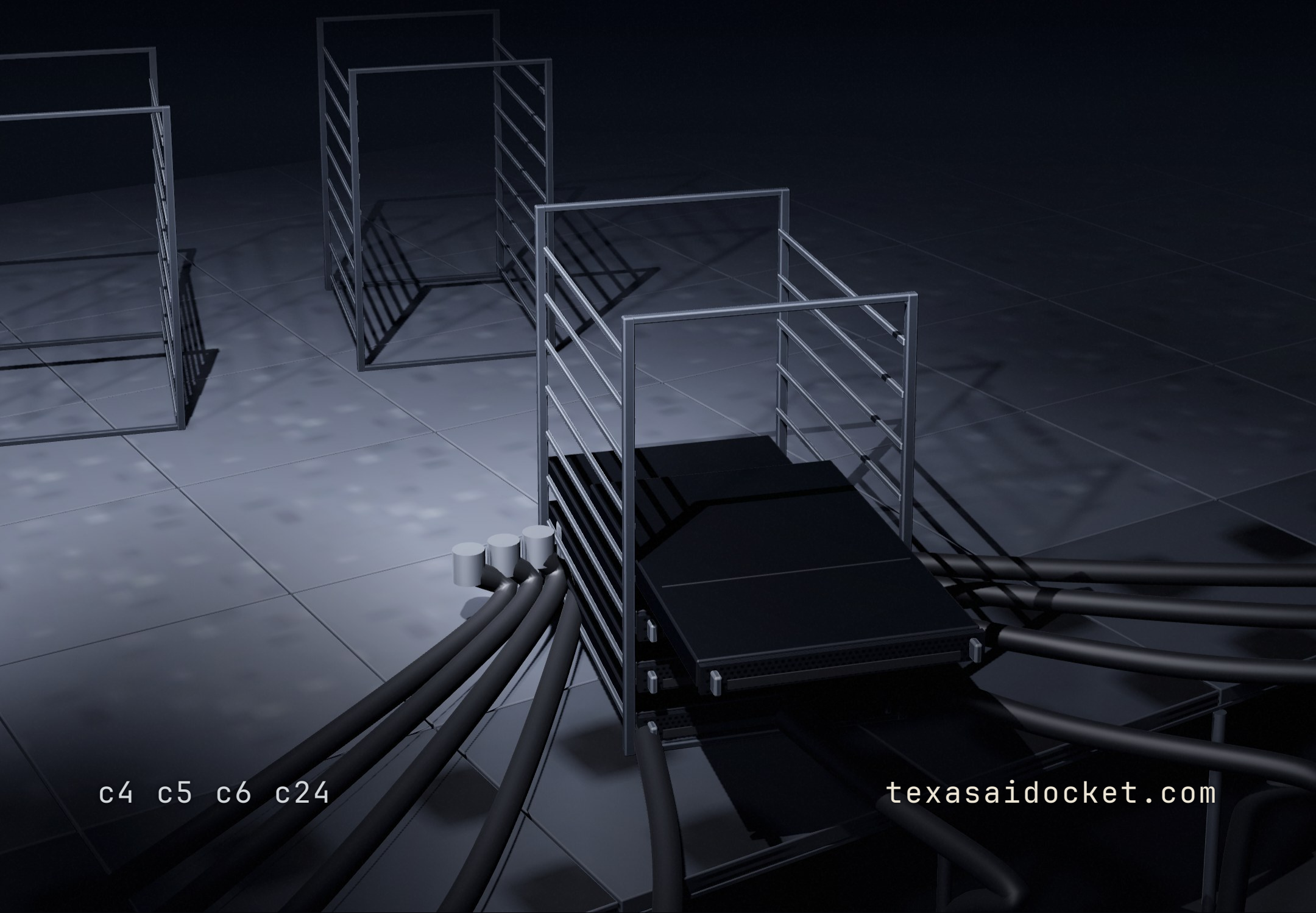
Vista has 860 compute nodes. TACC says the NVIDIA GH200 Grace Hopper Superchip is the processor for the majority. It pairs a Grace CPU with a Hopper GPU. One cell per node. His eight nodes are raised.



c9 c25 c26 c28

Horizon is in early operations

Frontera's successor is in its first phase. TACC's executive director says its power infrastructure is complete and an exabyte-capacity storage system is in place. Initial deployment of its AI-focused hardware is underway.



Frontera is dark. The work moved.

More than 130,000 U.S. researchers have used TACC's systems since 2001. For one of them, eight Vista GPU nodes gave "almost the same speed" as 64 on Frontera. An early Horizon user says it enters full production soon.



c1 c8 c9 c21 c22 c23

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